

OSCAR DE LA CERDA

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EDUCATION

Yale University, New Haven, CT

May 2028

- *ABET in Electrical Engineering*
- *Certificate in Programming*
- *Relevant Coursework:* Electromagnetism (PHYS 181), Introduction to Computer Engineering (EENG 201), Introduction to Communications and Control (ECE 202), Circuits and Systems Design (ECE 2031), Data Structures and Programming (CS 223), Digital Systems (ECE 348)

University of Iceland, Reykjavik, Iceland

June – July 2025

- *Relevant Coursework:* Renewable Energy Technology, Politics and Economics of Renewables

Tom C. Clark High School, San Antonio, Texas

June 2024

- *Awards:* Questbridge National College Match Finalist, 14 AP courses, National Hispanic Recognition from College Board, AP Scholar with Distinction from College Board

TECHNICAL SKILLS

- *Computer Languages:* C, C++, MATLAB, Racket, Verilog, PowerShell, Bash
- *Hardware and Electronics:* Circuit Design, Breadboard Electronics, Oscilloscopes, soldering, PCB assembly, Arduino, Sensor integration, Integrated circuitry, FPGA encoding
- *Tools:* Altium, LTSpice

EXPERIENCE

Quantum Network Research, *Undergraduate Researcher*

June 2026 - Now

- Developed and debugged Python scripts to automate FSM sweep sequences across a quad-cell position sensor, mapping boundary conditions and characterizing sensor response across the full field of view.
- Developing a closed-loop control system to stabilize laser pointing on a quad-cell sensor under simulated FSM disturbances, leveraging characterized sensor boundary conditions to inform controller design.

Yale Undergraduate Aerospace Association, *Sub team Lead*

November 2024 - Now

- Designed, simulated, and implemented an analog-to-digital converter (ADC) circuit to enable a Teensy microcontroller to process pressure, voltage, and current sensor data during pre-fuel testing
- Developed a complete avionics sensor-suite schematic integrating GPS, IMU, altimeter, MCU, and S-PLIG modules to improve system reliability and flight data acquisition
- Implemented SPI communication between the ADC and microcontroller for high-speed digital transfer of sensor measurements.

Yale Bulldogs Racing, *Sub team Member*

November 2024 – May 2025

- Assembled and soldered a high-voltage PCB for the team's Formula-style electric race car, supporting power-delivery reliability in competitive environments
- Authored technical documentation detailing internal electrical components to ensure compliance with safety and racing regulations

PROJECTS & LEADERSHIP EXPERIENCE

Piezoelectricity in Iceland, *Independent Project*

June – July 2025

- Modeled and evaluated piezoelectric energy-harvesting systems (cantilever beams, diaphragm configurations, harvesting eels) on Icelandic ships to offset greenhouse gas emissions
- Conducted modeling on MATLAB to quantify electrical energy that could be captured in North Atlantic Ocean waves